

MINOR PROJECT – 6

Leave Management System

Appsmith, PostgreSQL, Docker and AWS EC2 Deployment

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Project Leave Management System

Tools Used Appsmith, PostgreSQL, Docker, AWS EC2

Abstract

This report presents the design and implementation of a Leave Management System using Appsmith as the application interface, PostgreSQL for data storage, Docker for containerized deployment, and AWS EC2 as the cloud hosting environment. The system provides a leave request form, status tracking, search and filtering, and administrative approval or rejection of leave requests. Database records were verified through a PostgreSQL query, while the Docker containers and EC2 instance were verified as running.

1. Introduction

Managing employee leave requests through manual methods can make tracking, approval and record maintenance difficult. The Leave Management System provides a centralized interface in which employee leave information can be entered and stored in a PostgreSQL database. The application displays the stored requests in a table and provides controls for searching, approving and rejecting requests. The completed application is deployed using Docker containers and hosted on an AWS EC2 instance.

2. Objectives

- Create a web-based interface for submitting employee leave requests.
- Store leave request information in a PostgreSQL database.
- Display leave requests with employee details, dates, reason and current status.
- Provide Pending, Approved and Rejected status tracking.
- Provide search/filter functionality for leave records.
- Provide administrative Approve and Reject actions.
- Deploy the application and database using Docker containers.
- Host and verify the deployed system on AWS EC2.

3. Project Configuration

Component	Configuration
Application	Appsmith — Leave Management System
Database	PostgreSQL — leave_requests table
Application Container	Appsmith container
Database Container	PostgreSQL container
Application Port	80 mapped to the application container
Deployment	Docker on AWS EC2
Main Functions	Submit Leave Request, Search, Approve, Reject, Status Tracking
Verified Data	Employee name, email, leave type, start date, end date, reason, status

4. Application Interface and Leave Request Evidence

The application provides the main Leave Management System interface. The form contains fields for employee name, employee email, leave type, start date, end date and reason for leave. A Submit Leave Request button sends the request for processing. The page also displays the current leave status, search input, status counters and the leave records table.

The interface is titled "LEAVE MANAGEMENT SYSTEM". It features a form for submitting leave requests with the following fields:

- Employee Name* (text input)
- Employee Email* (text input)
- Leave Type* (dropdown menu)
- Start Date (date input, showing 2024-09-28 10:49)
- End Date (date input, showing 2024-09-28 10:49)
- Reason for Leave* (text input)

Below the form is a "Submit Leave Request" button. The "Leave Status" section shows "Pending". A search bar is labeled "Search Input" with a placeholder "Search employee email or leave type...".

Three status counters are displayed as horizontal bars:

- Pending: 0 (yellow bar)
- Approved: 5 (green bar)
- Rejected: 0 (red bar)

A table of leave records is shown below the counters. The table has columns: Employee No., Email, Leave Type, Start Date, End Date, Status, Reason, Action, and Reject. The data is as follows:

Employee No.	Email	Leave Type	Start Date	End Date	Status	Reason	Action	Reject
Arka	arka@example.com	CA B/L	28 September, 2024	29th Sep 2024	Approved	No check	Approved	Reject
Datta Mondal	datta@example.com	SICK	28 September, 2024	30th Sep 2024	Approved	personal work	Approved	Reject
Krutika	krutika@example.com	SICK	29 September, 2024	30th Sep 2024	Approved	Teaching prepared B...	Approved	Reject
Richa Kumar	richa@example.com	Sick Leave	29 September, 2024	30th Sep 2024	Approved	Not feeling well	Approved	Reject
Disha	disha@example.com	ANNUAL	28 September, 2024	29th Sep 2024	Approved	normalcy	Approved	Reject

Figure 1: Leave Management System interface with request form, status, search, counters and records table.

The interface provides a single place for entering a request and reviewing the stored leave records. The table includes employee details, leave type, dates, status, reason and action controls.

5. Leave Submission and Pending Status

A new leave request can be entered by providing the employee information, selecting a leave type, specifying the dates and entering the reason. After submission, the request is displayed with Pending status until it is processed by the administrator.

LEAVE MANAGEMENT SYSTEM

Employee name*

Sanchita

Employee Email*

sanchita@example.com

Leave Type*

Sick Leave

Start Date

2026-09-28 10:49

End Date

2026-09-28 10:49

Reason for Leave*

fever

Submit Leave Request

Leave Status

Pending

Search Input

Search employee, email or leave type...

Pending: 1

Approved: 5

Rejected: 0

Q Search...

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7 Records

Page 1 of 1

Employee Na...	Email	Leave Type	Start Date	End Date	Status	Reason	Action	Reject
Krishita	krishita@example...	CASUAL	28 September, 2026	28th Sep 2026	Pending	going out	Approve	Reject
Avia	avia@example.com	CASUAL	28 September, 2026	28th Sep 2026	Approved	to check	Approve	Reject
Dalia Mahata	dalia@example.com	SICK	28 September, 2026	30th Sep 2026	Approved	personal work	Approve	Reject
Krucks	krucks@example...	SICK	29 September, 2026	30th Sep 2026	Approved	Testing approve b...	Approve	Reject
Rahul Kumar	rahul@example.com	Sick Leave	29 September, 2026	30th Sep 2026	Approved	Not feeling well	Approve	Reject
Disha	disha@example.c...	ANNUAL	28 September, 2026	28th Sep 2026	Approved	normally	Approve	Reject

Figure 2: Leave request entered in the form and displayed with Pending status.

Employee Na...	Email	Leave Type	Start Date	End Date	Status	Reason	Action	Reject
Sanchita	sanchita@examl...	SICK	28 September, 2026	28th Sep 2026	Pending	fever	Approve	Reject
Krishita	krishita@example...	CASUAL	28 September, 2026	28th Sep 2026	Approved	going out	Approve	Reject
Avia	avia@example.com	CASUAL	28 September, 2026	28th Sep 2026	Approved	to check	Approve	Reject
Dalia Mahata	dalia@example.com	SICK	28 September, 2026	30th Sep 2026	Approved	personal work	Approve	Reject
Krucks	krucks@example...	SICK	29 September, 2026	30th Sep 2026	Approved	Testing approve b...	Approve	Reject
Rahul Kumar	rahul@example.com	Sick Leave	29 September, 2026	30th Sep 2026	Approved	Not feeling well	Approve	Reject

Figure 3: Leave records table showing employee information, leave type, dates and status.

6. Search and Filtering Evidence

The application includes a search field that can be used to locate leave records by employee information or leave type. The filtered result allows the administrator to inspect the matching request without manually reviewing every record.

The screenshot shows a dashboard titled 'Pending' with a search input field containing 'Avia'. Below the search field, there are three status summary boxes: 'Pending: 1' (yellow), 'Approved: 6' (green), and 'Rejected: 0' (red). Below these is a table with 9 columns: Employee No..., Email, Leave Type, Start Date, End Date, Status, Reason, Action, and Reject. The table has 1 record displayed, which is highlighted in purple. The record for 'Avia' has an email of 'avia@example.com', a leave type of 'CASUAL', start and end dates of '28 September, 2026', a status of 'Approved', a reason of 'to check', and action buttons for 'Approve' and 'Reject'. The table footer indicates '1 Records' and 'Page 1 of 1'.

Employee No...	Email	Leave Type	Start Date	End Date	Status	Reason	Action	Reject
Avia	avia@example.com	CASUAL	28 September, 2026	28th Sep 2026	Approved	to check	Approve	Reject

Figure 4: Search/filter operation displaying the matching leave record.

The evidence demonstrates that the dashboard can narrow the displayed records according to the entered search term while retaining the status summary.

7. Approval and Rejection Workflow

Each leave record contains Approve and Reject controls. Processing a request changes the status stored for that record. The screenshots demonstrate approved and rejected states in the leave table, allowing the administrator to track the outcome of each request.

Q Search...

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7 Records < Page 1 of 2 >

Employee Na...	Email	Leave Type	Start Date	End Date	Status	Reason	Action	Reject
Krishita	krishita@example....	CASUAL	28 September, 2026	28th Sep 2026	Approved	going out	Approve	Reject
Sanchita	sanchita@exampl...	SICK	28 September, 2026	28th Sep 2026	Approved	fever	Approve	Reject
Avia	avia@example.com	CASUAL	28 September, 2026	28th Sep 2026	Approved	to check	Approve	Reject
Dalia Mahata	dalia@example.com	SICK	28 September, 2026	30th Sep 2026	Approved	personal work	Approve	Reject
Krucks	krucks@example....	SICK	29 September, 2026	30th Sep 2026	Approved	Testing approve b...	Approve	Reject
Rahul Kumar	rahul@example.com	Sick Leave	29 September, 2026	30th Sep 2026	Approved	Not feeling well	Approve	Reject

Figure 5: Leave records showing the available Approve and Reject actions and processed statuses.

Q Search...

 Download

7 Records

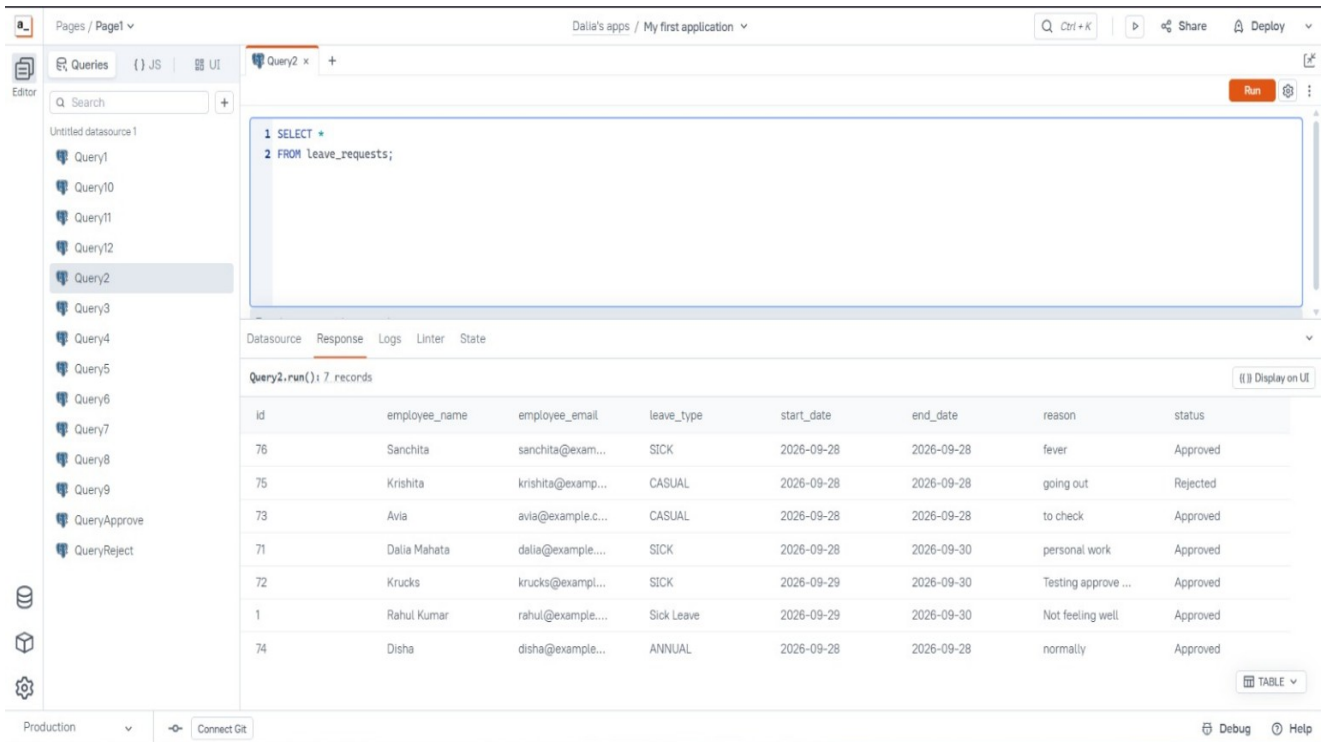
< Page 1 of 2 >

Employee Na...	Email	Leave Type	Start Date	End Date	Status	Reason	Action	Reject
Sanchita	sanchita@exampl...	SICK	28 September, 2026	28th Sep 2026	Approved	fever	Approve	Reject
Krishita	krishita@example....	CASUAL	28 September, 2026	28th Sep 2026	Rejected	going out	Approve	Reject
Avia	avia@example.com	CASUAL	28 September, 2026	28th Sep 2026	Approved	to check	Approve	Reject
Dalia Mahata	dalia@example.com	SICK	28 September, 2026	30th Sep 2026	Approved	personal work	Approve	Reject
Krucks	krucks@example....	SICK	29 September, 2026	30th Sep 2026	Approved	Testing approve b...	Approve	Reject
Rahul Kumar	rahul@example.com	Sick Leave	29 September, 2026	30th Sep 2026	Approved	Not feeling well	Approve	Reject

Figure 6: Updated leave table showing a rejected request together with approved requests.

8. Database Verification

The Leave Management System stores its records in PostgreSQL. The database evidence shows a SELECT query executed against the leave_requests table. The result contains the stored employee name, employee email, leave type, start date, end date, reason and status fields.



The screenshot shows a database query tool interface. The query editor contains the following SQL query:

```
1 SELECT *
2 FROM leave_requests;
```

The results pane shows the output of the query, displaying 7 records. The columns are: id, employee_name, employee_email, leave_type, start_date, end_date, reason, and status. The data is as follows:

id	employee_name	employee_email	leave_type	start_date	end_date	reason	status
76	Sanchita	sanchita@exam...	SICK	2026-09-28	2026-09-28	fever	Approved
75	Krishita	krishita@examp...	CASUAL	2026-09-28	2026-09-28	going out	Rejected
73	Avia	avia@example.c...	CASUAL	2026-09-28	2026-09-28	to check	Approved
71	Dalia Mahata	dalia@example....	SICK	2026-09-28	2026-09-30	personal work	Approved
72	Krucks	krucks@examp...	SICK	2026-09-29	2026-09-30	Testing approve ...	Approved
1	Rahul Kumar	rahul@example....	Sick Leave	2026-09-29	2026-09-30	Not feeling well	Approved
74	Disha	disha@example....	ANNUAL	2026-09-28	2026-09-28	normally	Approved

Figure 7: PostgreSQL leave_requests table queried to verify stored application records.

The database result confirms that the records displayed by the application are persisted in the PostgreSQL backend.

9. AWS EC2 Deployment Verification

The completed application was deployed on an AWS EC2 instance. The EC2 console evidence shows the instance in the Running state and identifies the deployed Leave Management System environment.

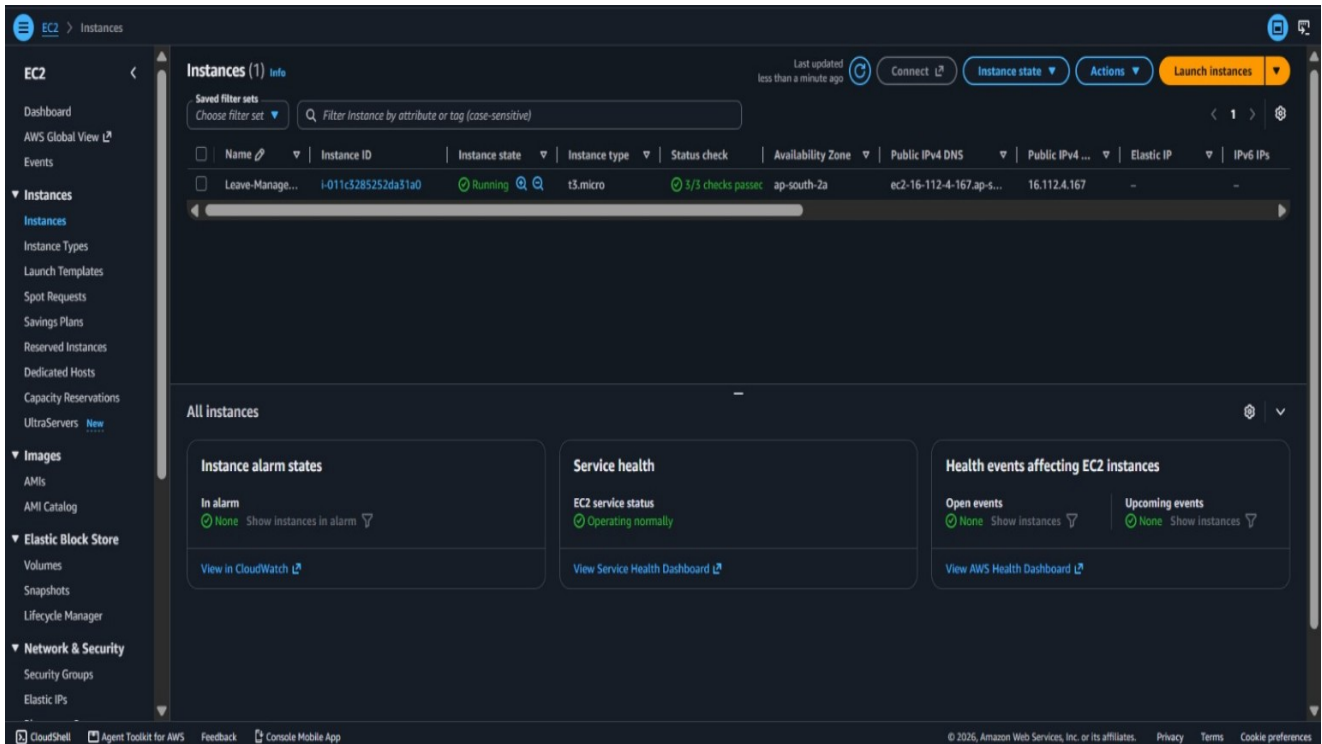


Figure 8: AWS EC2 instance running for the Leave Management System deployment.

The running EC2 instance provides the cloud environment used to host the containerized application.

11. Validation Summary

The Leave Management System was validated from the user interface through the database and deployment environment:

- Leave request form verified with employee, email, leave type, date and reason fields.
- Submitted request verified with Pending status.
- Approved and Rejected statuses verified in the leave records table.
- Pending, Approved and Rejected counters displayed on the dashboard.
- Search/filter operation verified using an employee record.
- PostgreSQL leave_requests table verified using a SELECT query.
- Appsmith and PostgreSQL Docker containers verified as running.
- AWS EC2 instance verified in the Running state.

12. Conclusion

This project successfully demonstrates a centralized Leave Management System using Appsmith, PostgreSQL, Docker and AWS EC2. The system supports leave submission, database storage, status tracking, searching, approval and rejection. PostgreSQL verification confirms persistent storage, while Docker and EC2 verification confirms the cloud deployment environment. The evidence demonstrates an end-to-end workflow from leave request entry to database storage and administrative processing.

13. References

- Appsmith documentation
- PostgreSQL documentation
- Docker documentation
- Amazon EC2 documentation

— End of Report —